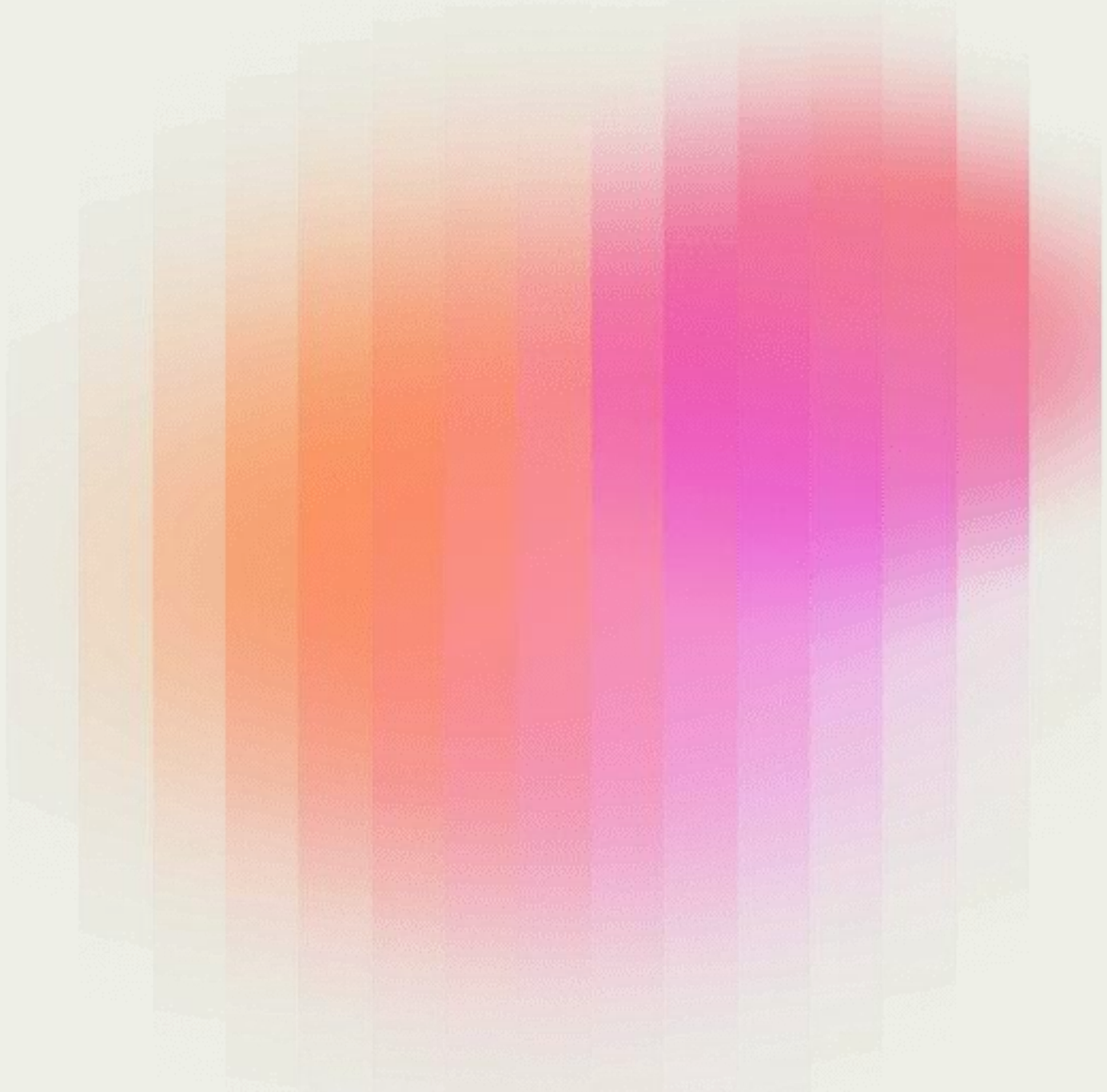
Presentation subtitle



Critical

1.1 What Is Philosophy?

Start Thinking about Philosophy

- Philosophy is a field that seeks knowledge of a wide range of things, each of which is wide-ranging—including nature, consciousness, morality, beauty, and social structure—and seeks great depth of knowledge.
- The word *philosophy* derives from ancient Greek, in which the philosopher is a lover or pursuer (*philia*) of wisdom (*sophia*).

1.1 What Is Philosophy?

Sages across Historical Traditions

- The earliest Greek philosophers were not known as philosophers; they were simply known as sages.
- Sages are people who are alleged to be wise. Sages existed in several historical traditions.
  - India
  - China
  - Africa
  - Greece

1.1 What Is Philosophy?

The Seven Sages of India (Saptarishi)

- The seven sages were considered partly real people and partly mythological: descendants of the gods.
- They were said to be authors of the Vedas, ancient Indian religious and philosophical texts.
- They were also considered spiritual guides who were involved in spiritual practices to control the mind and the body.



Figure 1.2 This painting, from the late eighteenth century, depicts the first man, Manu, guiding seven sages through floodwaters, with the aid of the king of serpents.
(credit: "Manu and Saptarishi" by unknown author/Wikimedia Commons, Public Domain)

1.1 What Is Philosophy?

Important Chinese philosophers

- Confucius and his followers advocated a society that was hierarchical, but in which the upper classes looked out for the lower classes. They also emphasized rituals.
- Daoists talked of a mysterious Dao, "the Way," of which everything is a part, including opposites. They also said that people should follow the Dao.



Figure 1.3 The Chinese philosopher and historian Han Feizi identified sages with technological discoveries. (credit: “Portrait of Han Fei” by unknown author/Wikimedia Commons, Public Domain)

1.1 What Is Philosophy?

1. Folk Sages

2. Philosophical Sages

Oruka was interested in sages who "showed a rational method of inquiry into the real natures of things," which the philosophical sages did, while the folk sages did not.

1.1 What Is Philosophy?

Tension in the Thought of Oruka's Sages

- The philosophers he looked at did express the beliefs that had been passed down in their cultures.
- However, they distanced themselves from their culture enough to give a critical analysis of the culture, and they sought rational justification for their beliefs.

1.1 What Is Philosophy?

The seven sages of Greece include:

- Thales of Miletus, who said that water is the fundamental element and that material things that move themselves have souls
- Solon of Athens, who restructured the government of Athens and abolished all debts, thus ending debt slavery

1.1 What Is Philosophy?

Greek Philosophers and Natural Philosophy (Science)

- Thales—said that water is the underlying matter causing things to be the way they are
- Xenophanes—tried to explain many facts of nature in terms of clouds
- Parmenides—tried to explain away all change as an illusion
- Pythagoras and the Pythagoreans—tried to explain facts about the world in terms of mathematical attributes: quantities

1.1 What Is Philosophy?

Greek Philosophers and Natural Philosophy (Science), continued

- The Atomists (Leucippus and Democritus)—tried to explain change in nature through the actions of "atoms" (indivisible particles)
- Socrates and Plato—tried to explain nature through matter and forms
- Aristotle—tried to explain nature through matter, forms, efficient causes (changing causes), and ends

1.1 What Is Philosophy?

Thales's Philosophical Dichotomy

Thales said that there are two kinds of elements:

1. Material elements, of which the fundamental one is water.
2. Elements with an internal source of motion: magnets and amber.

   These elements are those with souls.

Xenophanes

Xenophanes tried to explain several things in nature as operations of clouds: rainbows, the sun, the moon, and St. Elmo's Fire.

Some think that he claimed that all things are made of earth.

He questioned traditional beliefs about the gods.

1.1 What Is Philosophy?

Facts about the Pythagoreans

- They did much work in mathematics, including discovering the Pythagorean theorem.
- They saw the mathematical aspect of musical harmony and astronomy, and they concluded that nature was mathematical.
- They believed in an immortal soul in humans and in animals.
- They believed that the soul reincarnates.
- They lived in a community that had rules regarding diet, dress, and behavior.
- Women were included and had prominent roles.

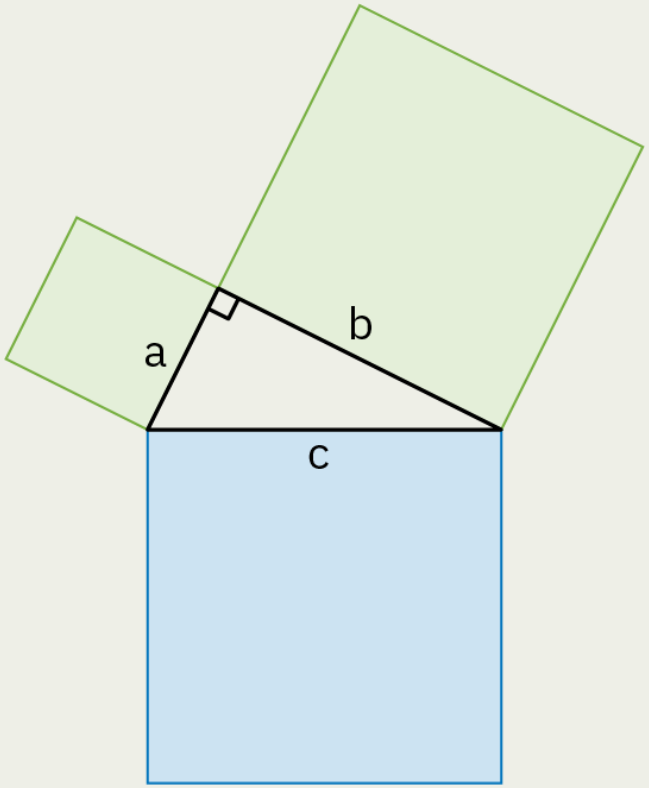


Figure 1.5 The Pythagorean Theorem describes the relationship between the sides of a right-angled triangle as demonstrated by the ancient Greek philosopher, Pythagoras.
(credit: Marianov/Wikimedia Commons, CC0)

1.1 What Is Philosophy?

Democritus's Philosophy

- The fundamental particles of matter are unchanging and are indivisible atoms.
- Perceivable qualities are not in the atoms but the product of convention.

1.1 What Is Philosophy?

How It All Hangs Together

- Taking a wide view: Wilfrid Sellars emphasizes that philosophy's goal is to understand a very wide range of topics—in fact, the widest possible range. That is to say, philosophers are committed to understanding everything insofar as it can be understood.

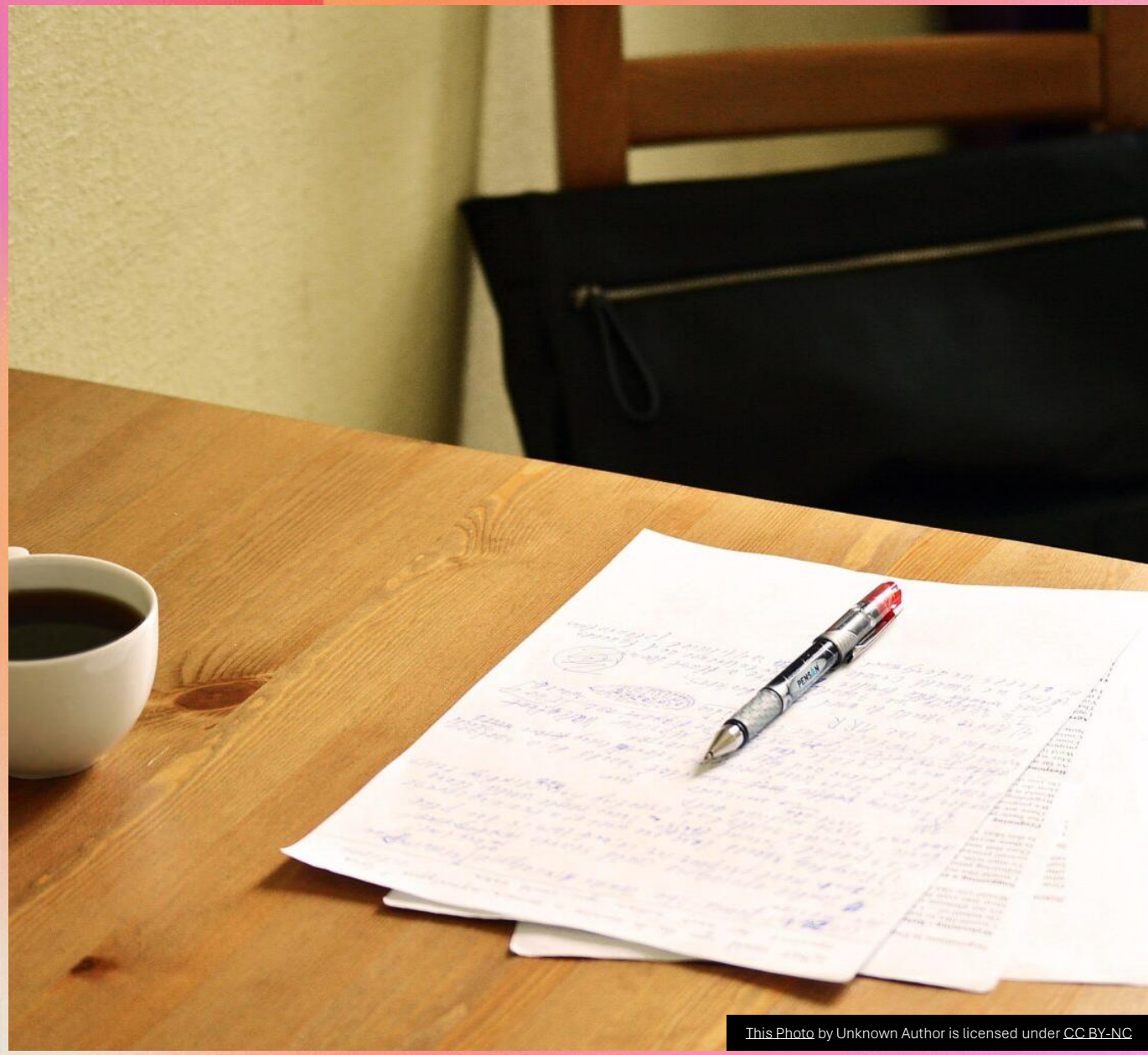
- Looking for connections: Sellars thinks that this philosophical skill is most clearly demonstrated when people try to understand the connection between the natural world as they experience it directly (the "manifest image") and the natural world as science explains it (the "scientific image").

2.1 The Brain Is an Inference Machine

- Evolution has adapted every part of humans to make survival of our genes more likely.
- Brains facilitate humans' survival and ability to find a partner and reproduce.
- Evolution made human brains into very effective inference machines.
- Nonetheless, the brain's conclusions may be incorrect.

Recap 24 Aug

1. Philosophy is a broad, cross-cultural inquiry into fundamental questions about reality, knowledge, values, and human life.
2. Critical thinking is the disciplined examination of how we form beliefs and make decisions.
3. Changing one's mind is not the only sign of critical thinking; understanding why we hold or revise beliefs is central.
4. Disciplinary boundaries such as science, engineering, and philosophy are historically recent; many problems require interdisciplinary thinking.



This Photo by Unknown Author is licensed under CC BY-NC


92NY
Neil deGrasse Tyson

26 Aug 2026

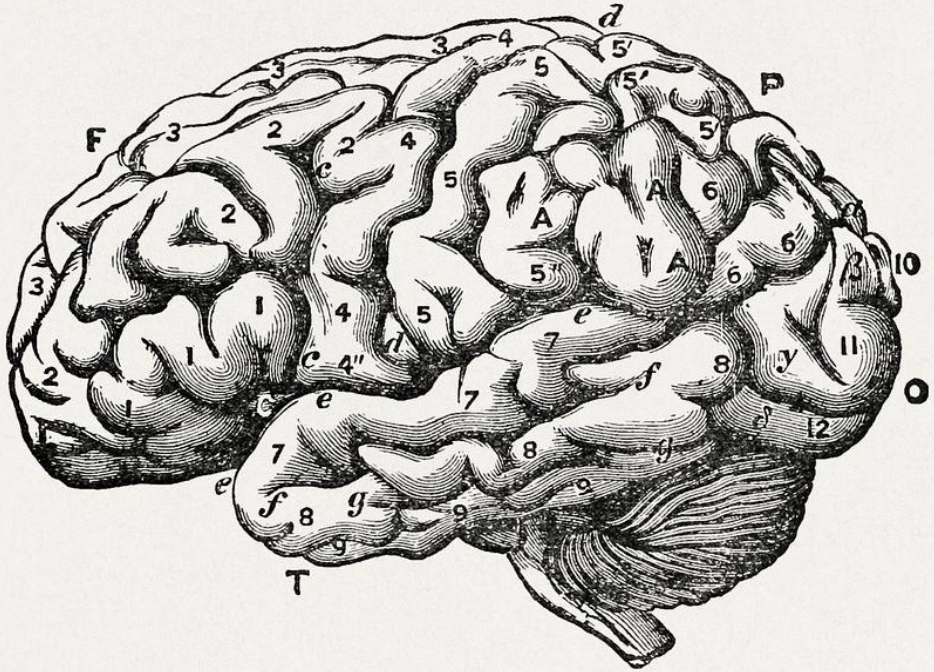
Philosophical Caveats about Brain Talk

We need to be cautious when we talk of brains and minds.

Some philosophers say that they are the same, but some do not.

Mental activity is not easily associated with any type of brain activity.

There seems to be more to the concept of minds than there is to the concept of brains.



2.1 The Brain Is an Inference Machine

Connection between the Mind and the Brain

- Even if the mind and the brain are different, they are closely connected
- Both psychology and cognitive science can help us understand how people think
- Critical thinking helps us understand how people *should* think.

2.1 The Brain Is an Inference Machine

Representation

- Many philosophers describe mental states as representations.
- Those philosophers would say that when people think about things, they represent those things.
- What is represented might be something present and real, something fictitious, something imagined in the future, or something remembered in the past.

2.1 The Brain Is an Inference Machine

Taking a Mental Picture

- You may think that in visual perception the brain passively takes a mental picture, with the eye functioning like a camera and the brain developing the picture.
- Problems:
  1. The picture is not completely filled in, in full focus, and uniformly colored.
  2. Your visual field is limited and hazy around the edges, with colors changing because of lighting, distance, movement, etc.

2.1 The Brain Is an Inference Machine

Inferences in Representation

- In representation, the brain makes inferences, even in the simple case of visual perception.
- Look at the checkered set of squares in the image. Are the horizontal lines parallel?
- In fact, the horizontal lines are parallel, but unless you look at the image from the side, it is impossible to visualize this.

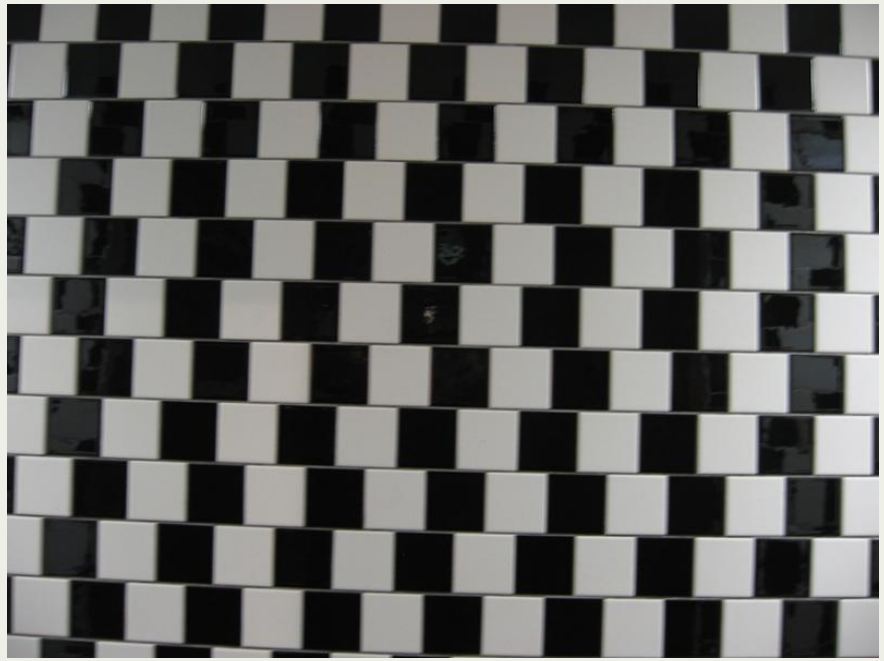


Figure 2.3 The horizontal lines on this grid are parallel, but unless you look at the image from the side, it is impossible to "see" this. This is one of many examples of common perceptual illusions.
(credit: "Optical Illusion" by Selena N. B. H., CC BY 2.0)

2.1 The Brain Is an Inference Machine

Further Problems with the Belief That the Mind Takes a Picture

1. Your brain is not passively capturing the world, like a camera, but is actively projecting the world so that it makes sense to you.
2. Where is the picture?
3. Who is viewing it?
4. The problem of optical illusions

2.1 The Brain Is an Inference Machine

Influence of Emotions on Reason

- Many people think that one must maintain skepticism of strong emotions.
- Neuroscientist Antonio Damasio (1994) said that philosophy is wrongly biased against emotions.
- Clinical practice has shown Damasio that reason and emotion cannot be separated; people reason better when they care about something.

2.1 The Brain Is an Inference Machine

Evolutionary Advantage of Shortcuts

- Deliberate thinking requires attention, and attention requires energy.
- Deliberate thinking makes energy demands on people, and so shortcuts in thought processes save energy.
- Humans have a biological tendency to avoid effortful thinking, and so it is evolutionarily wise to be lazy.

2.1 The Brain Is an Inference Machine

Heuristics

- Heuristics are mental shortcuts. They are mostly unconscious, automated, effortless, and efficient, but they are not always correct.
- Rational thinking or computation requires conscious effort and may be impossible without practice.
- People are forced to engage in effortful thinking when confronted with what is new and possibly dangerous, or something slightly outside of normal.



Selective Attention Test

2.1 The Brain Is an Inference Machine

The Use of Heuristics in Decision-Making

- Heuristics are often used to save time in thinking, sometimes automatically and unconsciously.
- However, they may produce errors. In this case, the brain is operating with cognitive bias.
- An example is substitution heuristic, when a person answers a difficult question by substituting an easier one to feel “cognitive ease.”

1.2 How Do Philosophers Arrive at Truth?

Sources of Evidence

- History
- Intuition
- Common sense
- Experimental philosophy
- Results from other disciplines

1.2 How Do Philosophers Arrive at Truth?

Topics in Conceptual Analysis

- Predicates—descriptive terms (things said about a subject)
- Descriptions—collections of predicates about a subject
- Enumerations—specify the nature of things (listing of the parts of a concept)
- Thought experiments—testing the consequences of a theory by analyzing the consequences of the concepts in it

1.2 How Do Philosophers Arrive at Truth?

Topics in Logic

- Arguments—collections of facts (premises) that provide evidence for a conclusion
- Explanations—facts from which a conclusion can be derived; start with the conclusion and look for the premises
- Coherence—consistency of premises

1.2 How Do Philosophers Arrive at Truth?

Trade-offs in philosophy

- In philosophy there is often a clash between a general principle that a person thinks is true and what seem to be applications of the principle, and so a person must give up or modify one or both.
- You may find that each one has its advantages, making the choice involve trade-offs, and so it may be a hard choice between them.

31 Aug 2026

Recap

- Critical thinking requires openness to new stimuli and willingness to revise understanding
- Context is essential for proper interpretation of any observation
- Disciplinary boundaries are useful but interdisciplinary knowledge is increasingly important
- Great questions are more important than great answers
- Observation is selective and never complete; be aware of your attentional biases
- Emotions play a complex role in decision-making, not always negative

5.1 Philosophical Methods for Discovering Truth

Logic

- Logic aims to formalize and describe reasoning processes used to arrive at claims.
- Logic is a study of both how people *do* reason and how they *ought* to reason.

5.1 Philosophical Methods for Discovering Truth

Dialectics and Philosophical Argumentation

- An argument in philosophy is a reasoned position—to argue is simply to offer a set of reasons in support of some conclusion.
- The participants in a dialectic enter into discourse with the aim of trading their poor or false beliefs for knowledge.
- Dialectics usually start with a question. An interlocutor offers an answer to the question, which is then scrutinized by all participants.

5.1 Philosophical Methods for Discovering Truth

Indian Dialectics and Debate

- The earliest known philosophical writings originate in India, as sections of the Vedas, which have been dated as far back as 1500 BCE.
- In the Upanishads, dialogues usually occur between two people: one who knows the truth and the other who seeks to know the truth.
- Classical Indian philosophical texts also refer to the occurrence of reasoned public debates.

*"seeker of truth" is the student who wants to understand questions about the self, death, and how to live (*dharma*), and whose inquiry culminates in recognizing this unity of self and ultimate reality rather than in mere doctrinal answers

5.1 Philosophical Methods for Discovering Truth

Greek Dialectics and Debate

- Socrates, perhaps the most famous ancient Greek philosopher, claimed that knowledge is opinion backed by argument.
- All of what is known about Socrates is through the writings of others—particularly the writings of Plato. Plato uses dialogues in all of his works, in which Socrates is almost always a participant.

5.1 Philosophical Methods for Discovering Truth

Aristotle: Dialectics and Logic

- If any ancient Greek philosopher most embodies the tie between dialectic and logic, it is Aristotle, who was a student of Plato.
- Aristotle's logic is the earliest formal, systematized account of inference known and was considered the most accurate and complete system until the late 19th century.
- Aristotle's system is taught in logic classes to this day.

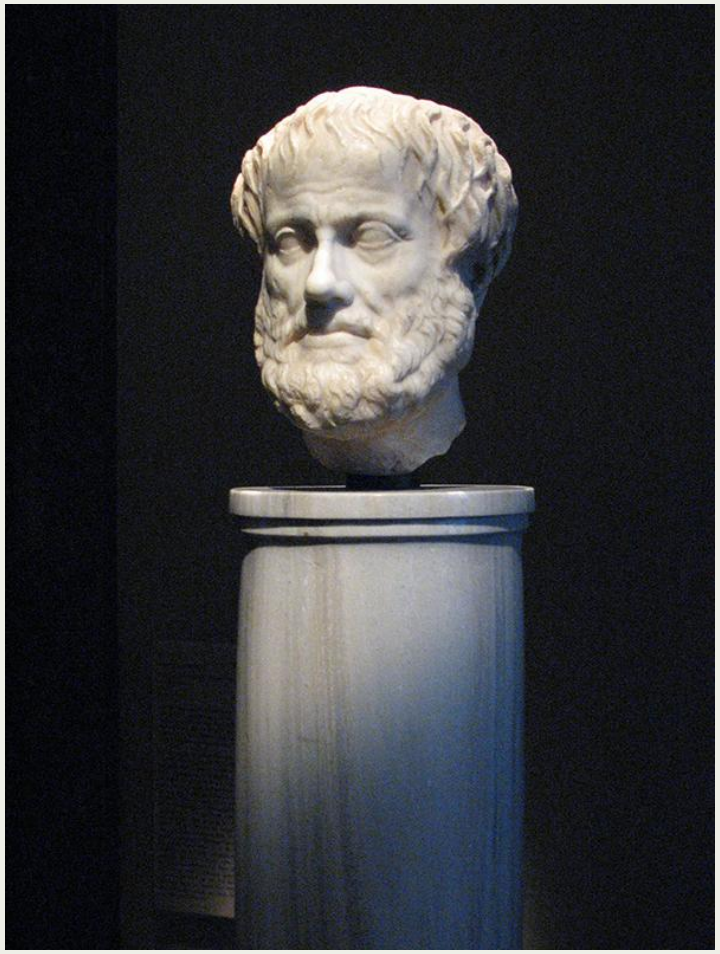


Figure 5.2 Roman copy in marble of a Greek bronze bust of Aristotle.
(credit: "Vienna 014" by Jeremy Thompson/flickr, CC BY 2.0)

5.1 Philosophical Methods for Discovering Truth

The Use of Reason to Discover Truth

- A **hypothesis** is a proposed explanation for an observed process or phenomenon. Human beings formulate hypotheses because they wish to answer specific questions about the world.
- **Reasoning** allows people to hypothesize, work out the consequences of their hypotheses, run thought experiments, assess the coherence of a set of beliefs, and generate plausible explanations of the world around them.
- People use **logic** and **reasoning** to test the truth of their suppositions.

Reasoning = making inferences in real situations

Logic = the systematic study of the principles that make those inferences valid or invalid, strong or weak.

Example 1 – Everyday reasoning (little explicit logic)

You come home and see your leftovers are gone; you live with one roommate. You think:

Premise: My food was in the fridge this morning.

Premise: The food is now gone.

Premise: Only my roommate and I have access.

Conclusion: My roommate probably ate my leftovers.

Here you're **reasoning**: drawing an inference based on everyday information and context.

You are not consciously applying formal rules—your thought is messy, probabilistic, and could be wrong (maybe someone else had a key). This is reasoning, and only *informally* logical.

Example 2 – A purely logical pattern

Consider the classic syllogism:

All mammals breathe air.

A dolphin is a mammal.

Therefore, a dolphin breathes air.philosophy.

Here we can describe the **logical form**:

If something is a mammal, then it breathes air.

A dolphin is a mammal.

Therefore, a dolphin breathes air.

A logician identifies this as an instance of *modus ponens* ("If P then Q; P; therefore Q"), and proves that whenever the premises are true, the conclusion must be true.philosophy.
That abstract study of the *form* of arguments, independent of topic (dolphins, Paris, Socrates, etc.), is **logic**. Reasoning is the mental act of connecting premises to conclusion; logic is the formal analysis of whether that connection is valid.

Example 3 – Reasoning with weak logic vs strong logic

Weak everyday reasoning:

Premises: Nikki saw a black cat; later she was fired.

Conclusion: Black cats bring bad luck.

This is a *reasoning move*—she's connecting events—but logically it's a bad inductive generalization with no real evidence.
Logic lets us label it as a fallacious pattern (post hoc/generalization) and explain why the conclusion doesn't follow.

Strong, logically good reasoning:

Premise: Every person who lives in Quebec lives in Canada.

Premise: Everyone in Canada lives in North America.

Conclusion: Every person who lives in Quebec lives in North America.

Here the reasoning fits a valid logical form (if all A are B and all B are C, then all A are C). Logic tells us: "This form guarantees the conclusion if the premises are true."

5.1 Philosophical Methods for Discovering Truth

Laws of Logic: The Law of Non-Contradiction

- **A statement** is a sentence with truth value, meaning that the statement must be true or false. Statements are declarative sentences. Sometimes philosophers use the term "**proposition**" instead of "statement."
- Contradictory propositions cannot be true *in the same sense, at the same time*.

5.1 Philosophical Methods for Discovering Truth

Laws of Logic: The Law of the Excluded Middle

- The law of the excluded middle states that for any statement, either that statement is true, or its negation is true.
- If the only available options for truth-bearing statements are that they are true or false, and if a statement and its negation cannot both be true at the same time, then one of the statements must be true and the other must be false.

5.1 Philosophical Methods for Discovering Truth

Normativity in Logic

- Normativity is the assumption that certain actions, beliefs, or other mental states are good and ought to be pursued or realized.
- Normativity implies a standard (a norm) to which one ought to conform.
- While ethics is normative in the realm of actions and behavior, logic is normative in the realm of reasoning.

5.2 Logical Statements

Conditionals

- A conditional is most commonly expressed as an “if–then” statement.
  - Examples: “If you eat your meat, then you can have some pudding”; “if that animal is a dog, then it is a mammal.”
- All conditionals include two components—that which follows the “if” and that which follows the “then.”
- Whatever follows “if” is called the antecedent; whatever follows “then” is called the consequent.

5.2 Logical Statements

Conditionals

- Necessary and sufficient conditions
  - All conditionals express two relations or conditions: those that are necessary and those that are sufficient.
  - If something is sufficient, it is always sufficient for something else. And if something is necessary, it is always necessary for something else.
    - For example, 120 credit hours are required for graduation, so 120 credit hours is necessary if you expect to graduate.

5.2 Logical Statements

Conditionals

- Counterexamples
  - Sometimes people disagree with conditionals by using counterexamples.
  - Counterexamples are important for testing the truth of propositions.

5.2 Logical Statements

Universal Statements

- Universal affirmative statements take two groups of things and claim all members of the first group are also members of the second group: “All A are B.”
- Universal affirmative statements are an alternative expression of (and logically equivalent to) a conditional.
- Universal affirmative statements also can be disproven using counterexamples.

5.3 Arguments

An argument in philosophy is simply a set of reasons offered in support of some conclusion.

Arguments have two components: the conclusion and the reasons offered to support it. The conclusion is what an arguer wants people to believe. The reasons offered are called premises.

Getting to the Premises

The first step in understanding an argument is to identify the conclusion.

“What is this person trying to make me accept?”

After you have identified the conclusion, try to summarize it as best you can.

Then, identify the premises or evidence the arguer offers in support of that conclusion.

Premise Indicators and Conclusion Indicators

Purpose	Specific Examples
Conclusion indicator words and phrases	therefore, hence, so, thus, consequently, accordingly, as a result, it follows that, it entails that, we can conclude, for this reason, it must be that, it has to be that
Premise indicator words and phrases	given that, since, because, for, in that, for the reason that, in as much as, as indicated by, seeing how, seeing that, it follows from, owing to, it may be inferred from

Table 5.1 Navigating an Argument

The Difference between Truth and Logic

- Logical analysis
  - Logical analysis ascertains whether the premises of an argument support the conclusion.
- Truth analysis
  - Truth analysis is the determination of whether statements are correct or accurate.

5.4 Types of Inferences: Deductive Reasoning

1. Deductive inferences are inferences arrived at through deduction (deductive reasoning).
2. Deductive inferences can guarantee truth because they focus on the structure of arguments.

5.4 Types of Inferences: Deductive Reasoning

1. Valid deductive inferences

1. A good deductive inference is called a valid inference, meaning its structure guarantees the truth of its conclusion given the truth of the premises.
2. If it is impossible for the conclusion to be false when the premises are assumed to be true, then the argument is valid.

Deductive Reasoning

Types of valid deductive inferences

Disjunctive syllogism

X or Y.
Not Y.
Therefore, X.

Modus ponens

If X, then Y.
X.
Therefore, Y.

Modus tollens

If X, then Y.

Not Y.

Therefore, not X.

5.4 Types of Inferences: Deductive Reasoning

Invalid deductive inferences

A bad deductive inference is called an invalid inference.

In invalid inferences, the structure does not guarantee the truth of the conclusion—that is to say, even if the premises are true, the conclusion may be false. This does not mean that the conclusion must be false, but that one simply cannot know whether the conclusion is true or false.


She has got a wart.

P1: All witches are things that can be burned.

P2: All things that can burn are made of wood.

C: Therefore, all witches are made of wood.

1. P1: All witches are things that can be burned.
2. P2: All things that can burn are made of wood.
3. C: Therefore, all witches are made of wood.

P1: All things made of wood float.

P2: All things that weigh the same as a duck float.

C: Therefore, all things that weigh the same as a duck are made of wood.

P1: All things made of wood float.

P2: All things that weigh the same as a duck float.

C: Therefore, all things that weigh the same as a duck are made of wood.

fallacy of the undistributed middle: just because two groups share a property (floating) doesn't mean they're the same group

P1: All witches are made of wood.

P2: This woman is made of wood

(because she weighs the same as a duck).

C: Therefore, this woman is a witch.

P1: All witches are made of wood.

P2: This woman is made of wood

(because she weighs the same as a duck).

C: Therefore, this woman is a witch.

Types of invalid deductive inferences

Affirming the consequent

If X, then Y.

Y.

Therefore, X.

Denying the antecedent

If X, then Y.

Not X.

Therefore, not Y.

5.4 Types of Inferences

Deductive Reasoning

- The Wason Selection Task



5.4 Types of Inferences

Inductive Inferences

- When you reason inductively, you gather evidence using your experience of the world and draw general conclusions based on that experience.
- Inductive reasoning (induction) is also the process by which you use general beliefs you have about the world to create beliefs about your particular experiences or about what to expect in the future.

5.4 Types of Inferences

Inductive Inferences

- Reasoning from specific instances to generalities
  - All evidence is gathered from particular instances, but the conclusion is a general one.

P: Every time I've touched a hot stove, I got burned.

C: Hot stoves burn skin.

5.4 Types of Inferences

Inductive Inferences

- Reasoning from specific instances to generalities
  - All evidence is gathered from particular instances, but the conclusion is a general one.

P: Three people from X country were rude

C: People from X country are rude

Pattern:

Observed: Instance 1, Instance 2, Instance 3…

Conclude: “All / Most X are Y.”

Risk:

Small or biased sample

Ignoring counterexamples

Confusing “some” with “all”

5.4 Types of Inferences

Inductive Inferences

- Reasoning from generalities to specific instances
  - You can also reason from an accepted generalization to particular instances.

Generality:
“All high-voltage equipment must be treated as live until proven otherwise.”
Specific instance:
A technician approaches an unfamiliar electrical panel.
Deductive move:
“This is high-voltage equipment, so I must treat it as live and follow safety procedures.”

5.4 Types of Inferences

Inductive Inferences

- Reasoning from generalities to specific instances
  - You can also reason from an accepted generalization to particular instances.

Generality:
“Technology in class distracts students and harms learning.”
Specific instance:
One student typing notes on a laptop, or using a dictionary app.
Deductive move:
“Therefore, this student is distracted and not learning.”

5.4 Types of Inferences

Inductive Inferences

- Reasoning from past to future
  - Based on ample experience of the past, you have a basis for prediction.

Works:
Past: "Every year in this region, July and August are the hottest months."
Future prediction: "Next July and August will also be the hottest months."
Why it works: Climate patterns are stable over human timescales; this is a strong, well-supported projection.

Fails:
Past: "It has not rained for the last 10 days."
Future prediction: "It will not rain tomorrow either."
Why it fails: Short-term weather is highly variable; a dry spell doesn't guarantee continued dryness.

5.4 Types of Inferences

Strong Inductive Inferences

- A *strong* inductive inference is one where, if the evidence offered is true, then the conclusion is probably true.

Observed instances:
Crash data over many years shows that people wearing seatbelts are far more likely to survive serious accidents than those not wearing them.

Inductive conclusion:
“Wearing a seatbelt greatly increases your chances of surviving a car crash.”

5.4 Types of Inferences

- A *weak* inductive inference is one where, if the evidence offered is true, the conclusion is not probably true.

Observed instances:
While traveling, someone experiences two openly rude or discriminatory interactions with people from a particular country. Both incidents are emotionally intense and memorable.

Inductive conclusion:
“People from that country are racist / hostile.”

Why it’s weak:
Extremely small sample relative to an entire population.
Emotional salience makes these cases feel “typical.”
Ignores vast variation and positive interactions.

5.4 Types of Inferences

Abductive Reasoning

- In abductive reasoning, the conclusion is meant to explain the evidence offered in the premises. In induction the premises explain the conclusion, but in abduction the conclusion explains the premises.
- Inference to the best explanation
  - Start with a set of data and attempt to come up with some unifying hypothesis that can best explain the existence of those data.

Observation:
A student who used to attend regularly and participate suddenly stops showing up, without explanation.
Possible explanations:
A) They lost interest in the course.
B) They're facing personal or health issues.
C) They're working extra hours and can't manage the schedule.

Abductive move (by staff):
"The best explanation is that something personal or practical is going on; we should reach out and offer support."

5.4 Types of Inferences

Abductive Reasoning

- Explanatory virtues
  - Explanatory virtues are aspects of an explanation that generally make it strong.
  - A good hypothesis should be *explanatory*, *simple*, and *conservative* and have *depth*.

Observation:
One area has far more trees, parks, and shaded streets than nearby neighbourhoods.

Explanation A (structural, historically grounded):
“This area was historically wealthier and had stronger lobbying power; municipal planting programmes prioritized it.”

Explanation B (individualistic, narrow):
“People here just care more about nature; they planted trees themselves.”

Explanatory virtues at stake:
Scope: A explains systematic city-wide patterns (many cities show similar inequalities).
Fit with data: A aligns with urban planning records, budget allocations, and historical maps.
Depth: A connects to broader theories of inequality and policy.

5.4 Types of Inferences

Abductive Reasoning

- Extraordinary claims require extraordinary evidence.
  - A novel claim that disrupts accepted knowledge will need more evidence to make it credible than a claim that already aligns with accepted knowledge.

Claim	Evidence that may be enough	Evidence needed before acceptance
“I saw a fox near campus.”	A credible photo or several eyewitnesses	Ordinary evidence is likely adequate
“A fox spoke to me in English.”	A personal story or a video clip	High-quality recording, independent verification, and serious exclusion of trickery or error

5.4 Types of Inferences

Abductive Reasoning

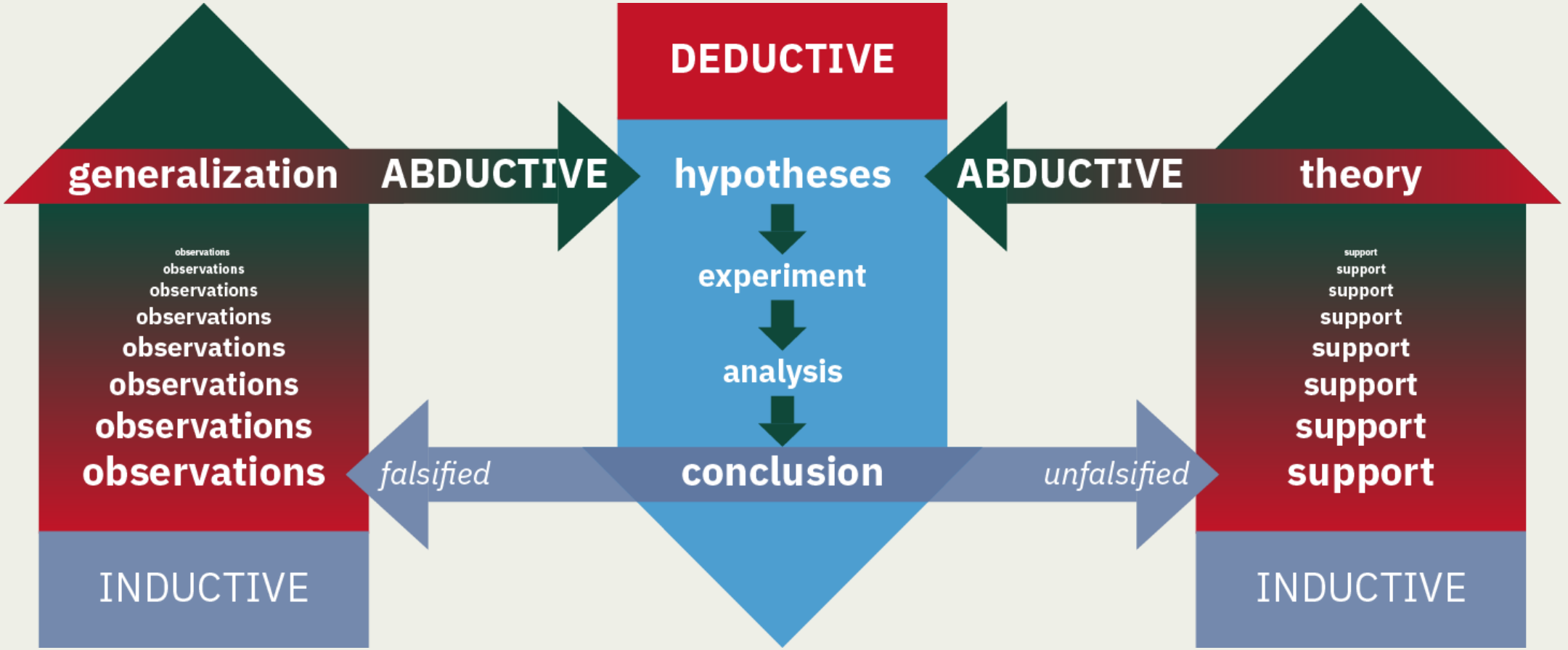


Figure 5.6 Induction in the Scientific Method
(attribution: Copyright Rice University, OpenStax, under CC BY 4.0 license)

Identify the type of reasoning

All third-year students are either in Energy Engineering or Computer Science. This student is not in Computer Science. Therefore, this student is in Energy Engineering.

Identify the type of reasoning

Over the last five years, every time the library has extended its opening hours during exams, usage has increased. Therefore, if the library extends hours during the next exam period, usage will probably increase again.

Identify the type of reasoning

All submissions received after the deadline without an approved extension incur a late penalty. This submission was received after the deadline and has no approved extension. Therefore, this submission incurs a late penalty.

Identify the type of reasoning

In several different courses, students who attend tutorials regularly and complete practice problems tend to score higher on average than those who do not. Therefore, regular tutorial attendance and practice probably improve performance.

Identify the type of reasoning

A friend who usually replies quickly has not responded for three days. The best explanation is that they are away or have limited internet access, rather than being angry with me.

Identify the type of reasoning

If a course requires 80% attendance to be eligible for the exam, and this student has only 65% attendance, then this student is not eligible for the exam.

Identify the type of reasoning

Every July for the past ten years has been hotter than June in this region. Therefore, next July will probably be hotter than next June.

Identify the type of reasoning

The building lost power for ten minutes, then it came back on. The most plausible explanation is a temporary grid fault, rather than every appliance in the building failing at exactly the same time.

Identify the type of reasoning

All high-voltage equipment must be treated as live until proven otherwise. This panel is high-voltage equipment. Therefore, this panel must be treated as live.

5.5 Informal Fallacies

- When the form of an argument is problematic, it is called a formal fallacy.

P1: If it rained, the street is wet.
P2: The street is wet.
C: Therefore, it rained.

- If the reasons are not adequate support for the conclusion, then the reasoner commits an informal fallacy.

5.5 Informal Fallacies

General Category	Specific Type	Description	
Fallacies of relevance— rely on evidence that is not relevant for logically establishing a conclusion			
	Appeal to emotion	Appeals to feelings (whether positive or negative) rather than discussing the merits of an idea or proposal	Everyone believes this diet works, so it must be effective.
	Ad hominem attack	Argues against someone's idea or suggestion by attacking the individual personally rather than pointing out problems with the idea or suggestion	Don't listen to her argument about climate; she's not a scientist.

5.5 Informal Fallacies

General Category	Specific Type	Description	Example
Fallacies of weak induction—rely on evidence or reasons that are too weak to firmly establish a conclusion			
	Hasty generalization	Draws a conclusion using too little evidence to support the conclusion	**Observed instances:** Two local startups you know of shut down after a year. Both struggled with funding and customers. **General claim:** "No one succeeds with startups in this city/country; it's impossible here."
	Biased sample	Draws a conclusion using evidence that is biased in some way	A politician posts a poll on Instagram: "Do you support Policy X?" Followers vote; 80% say yes. **General claim:** "Most people in the country support Policy X."
	Appeal to ignorance	Relies on the lack of knowledge or evidence for a thing (people's ignorance of it) to draw a definite conclusion about that thing	There's no official record or news report about this incident. Therefore, it never happened.

5.5 Informal Fallacies

General Category	Specific Type	Description	
Fallacies of unwarranted assumption—rely on information or beliefs that require further justification			
	False dichotomy	A limited number of possibilities are assumed to be the only available options	You are either born talented at music, or you will never be good at it.
	Begging the question	Either assumes the truth of a conclusion in the course of trying to prove it or assumes the truth of a contentious claim	We must trust this news source because it is the most trustworthy

5.5 Informal Fallacies

General Category	Specific Type	Description	
Fallacies of diversion—rely on attempts to distract the attention of the audience from the argument at hand			
	Strawman	Utilizes a weaker version of the position being argued against in order to make the position easier to defeat	"You say we should reduce military spending; so you want to leave the country defenseless."
	Red herring	Ignores the opponent's position and simply changes the subject	Q: "Did the minister use public money for personal travel?" R: "What people really care about is national security—and this government has kept the country safe."

No one has proved that this new app collects unnecessary data, so it must be safe to use.

Either we ban all AI tools from the university, or academic standards will collapse completely.

This policy is fair because it treats everyone fairly.

Identify

I asked five friends in my hostel what they thought about the new fees. All of them said the fees are reasonable. Therefore, most students on campus find the fees reasonable.

Identify

All engineers are good at mathematics. All physicists are good at mathematics. Therefore, all physicists are engineers.

Why are you criticising the university’s ranking? You should be talking about how beautiful our new sports complex is.

Identify

My neighbour used an AI tool to write part of his essay and got a high grade. AI tools clearly guarantee top marks for anyone who uses them.

Identify

If a student attends all lectures, they will pass the course. This student did not attend all lectures. Therefore, this student will not pass the course.

Identify

Three students in my tutorial said the library is always too crowded. So the library is clearly inadequate for the whole student body.

We cannot say that online exams are unreliable. After all, there is no solid evidence that they are completely secure.

A student posted online that an AI tool gave them an incorrect historical date. Two classmates then reported that it had produced weak summaries for their readings. It is obvious that AI is useless for serious education. The university must either ban every AI tool immediately or accept the end of genuine learning. In any case, no one has proved that a total ban would harm students, so there is no reason not to impose one. And critics of the ban should remember that the university has recently invested in an impressive new sports complex.

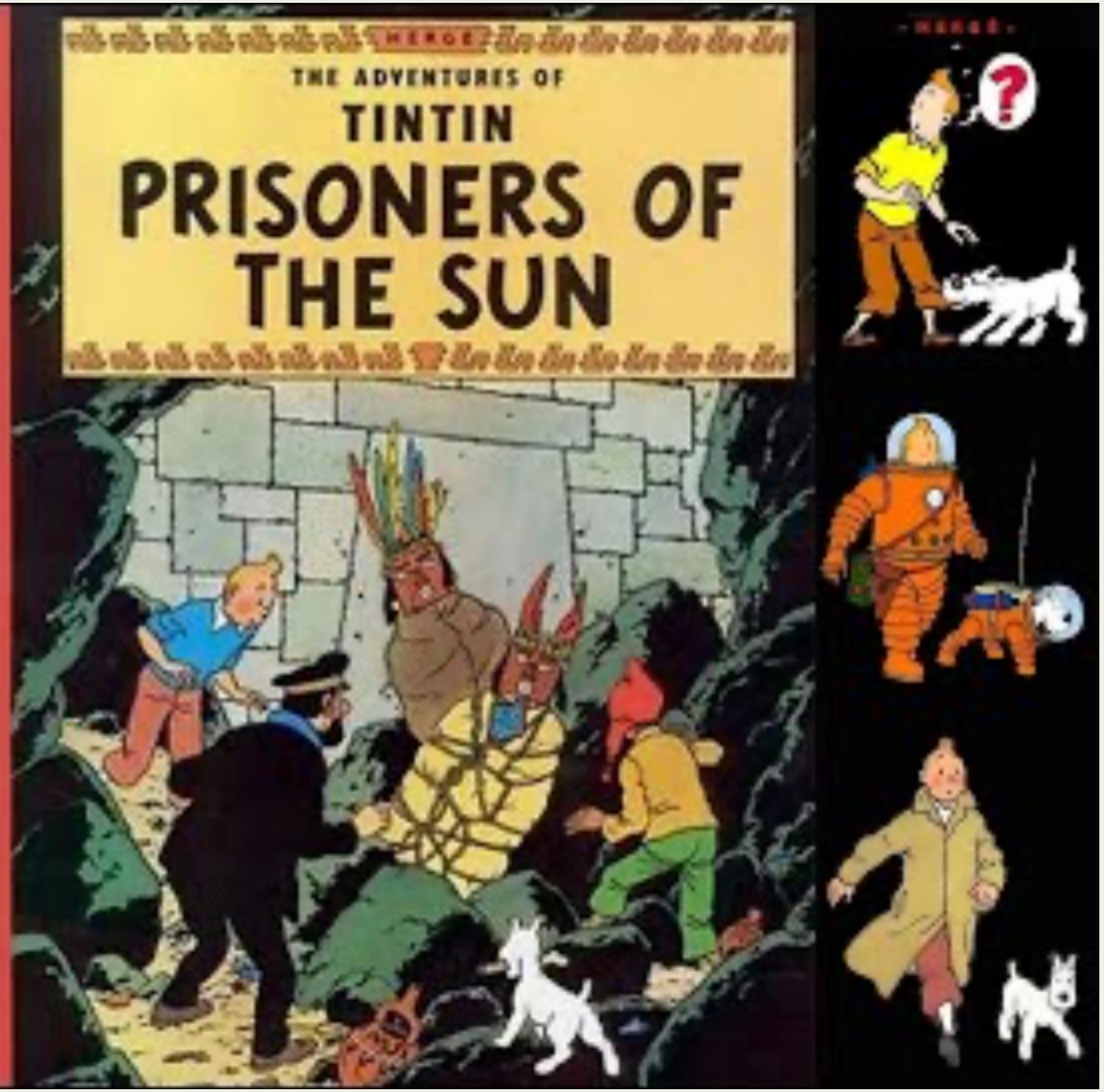
Identify:
hasty generalisation.
false dichotomy.
appeal to ignorance.
red herring.

කතා අංක 12

හිරු දෙවොලේ සිරකරුවෝ

(ඉංග්‍රීසි බසින් හඬ කැවූ)





CRITICAL THINKING · SEPTEMBER 16

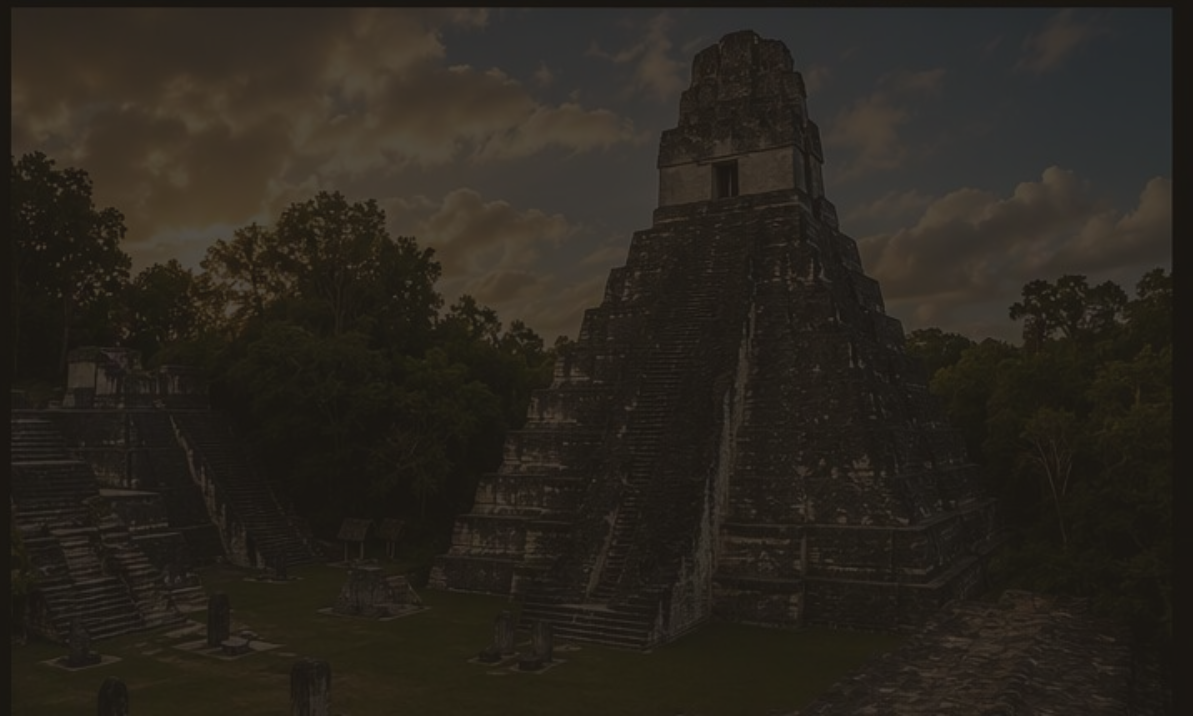
Philosophy before Europe

The Americas and Africa · Chapter 3

Who counts as a philosopher, and what counts as philosophical knowledge?

From Tintin's eclipse to Maya, Aztec, and Yoruba thought





PART I · THE AMERICAS

Mesoamerican Context: Maya & Aztec

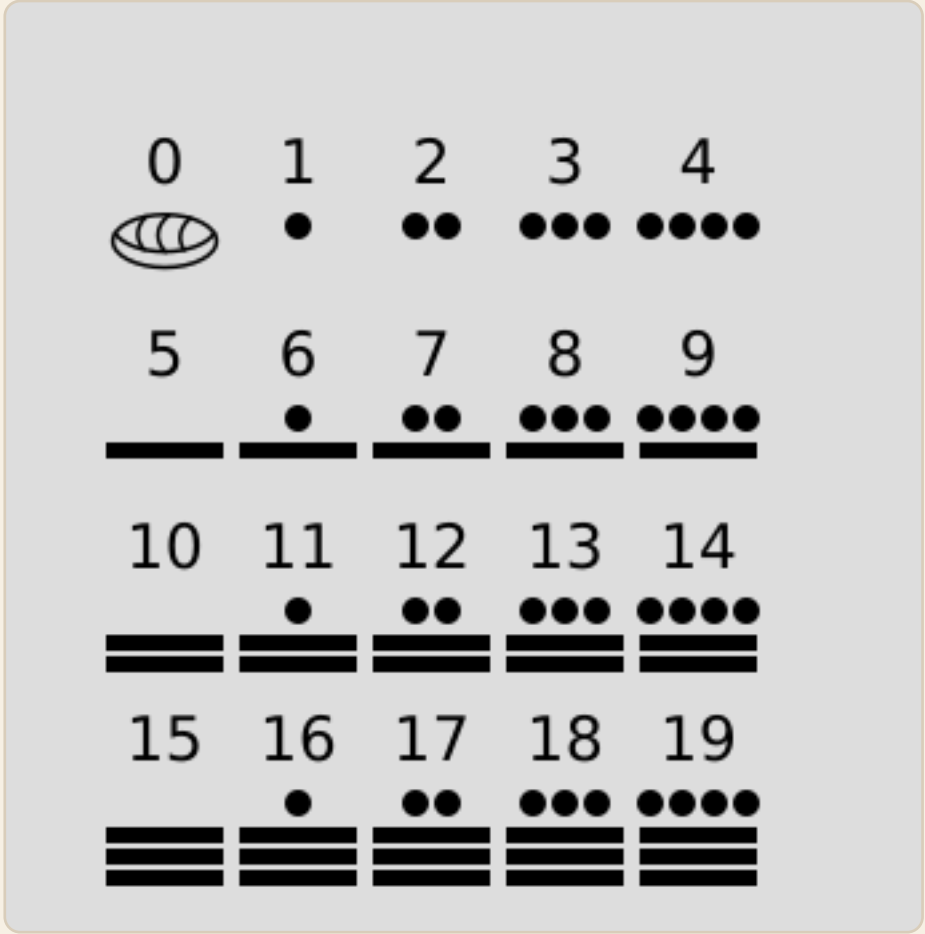
- Southern Mexico → Guatemala → northern Belize
- Writing combines pictographic, hieroglyphic, and phonetic elements
- Media include stone, pottery, sculpture, and bark-paper codices
- Long historical arc: villages → city-states → Classic Maya peak



PART I · THE AMERICAS

Maya: Mathematics, Astronomy, Calendar

- Positional number system and a sign for zero
- Systematic observation supported eclipse prediction
- Venus positions aligned with ritual and political action
- 260-day sacred count + 365-day solar year → 52-year Calendar Round



PART I · THE AMERICAS

Maya: Time and K’in

- Time is lived: boredom lengthens it; concentration compresses it
- Awe intensifies awareness of the present and cosmic forces
- K’in links sun, energy, time, sacred force, and political authority
- Prediction is both knowledge and legitimacy



PART I · THE AMERICAS

Aztec Metaphysics: What Is Teotl?

- A single sacred power or process underlying everything
- Pantheistic: divinity is present throughout reality
- Monist: reality is one dynamic “stuff”
- Immanent and amoral: not a distant lawgiver or judge

PART I · THE AMERICAS

Teotl as Process, Not Thing

- Warp, weft, and weaving are aspects of one unfolding activity
- Reality is dynamic rather than a collection of fixed substances
- Purposeful tendencies unfold within recurring cycles
- Mind/matter and animate/inanimate are guises of one motion

PART I · THE AMERICAS

“The Earth Is Slippery”

- Existence is unstable, changing, and difficult to stand securely within
- The ethical problem is practical: how can a person maintain balance?
- Character and disciplined practice create steadiness amid flux



PART I · THE AMERICAS

Neltiliztli: Well-Rootedness

- Truth is not correct belief alone
- To know is to be stably founded in reality
- Understanding, action, and character belong together
- A rooted life grows toward flourishing

African, Indian, and Chinese Philosophy

PHILOSOPHY WITHOUT AND BEYOND THE WRITTEN CANON

Africa → India → China



Recap

INCA / ANDEAN

Tintin's Prisoners of the Sun is primarily an Inca/Andean story.

DISTINCT CIVILIZATIONS

Maya, Aztec, and Inca refer to different civilizations, regions, and periods.

THE BROADER POINT

The critique of Eurocentric representation still stands.

Separate contexts

Compare carefully

Avoid collapse

Going forward, historical contexts remain distinct even when analytical questions cross regions.

Five questions, three traditions

03
QUESTIONS

1. What is the human problem?
2. What is reality like?
3. How can we know?
4. How should we live?
5. What is the goal of human life?

Use the same questions for Africa, India, and China.

Our path today

01

AFRICA

Oral traditions • colonial erasure • ubuntu • Yoruba thought

02

INDIA

Vedic orality • Buddhism • Jainism • authority

03

CHINA

Confucianism • Daoism • order • harmony

THROUGH-LINE

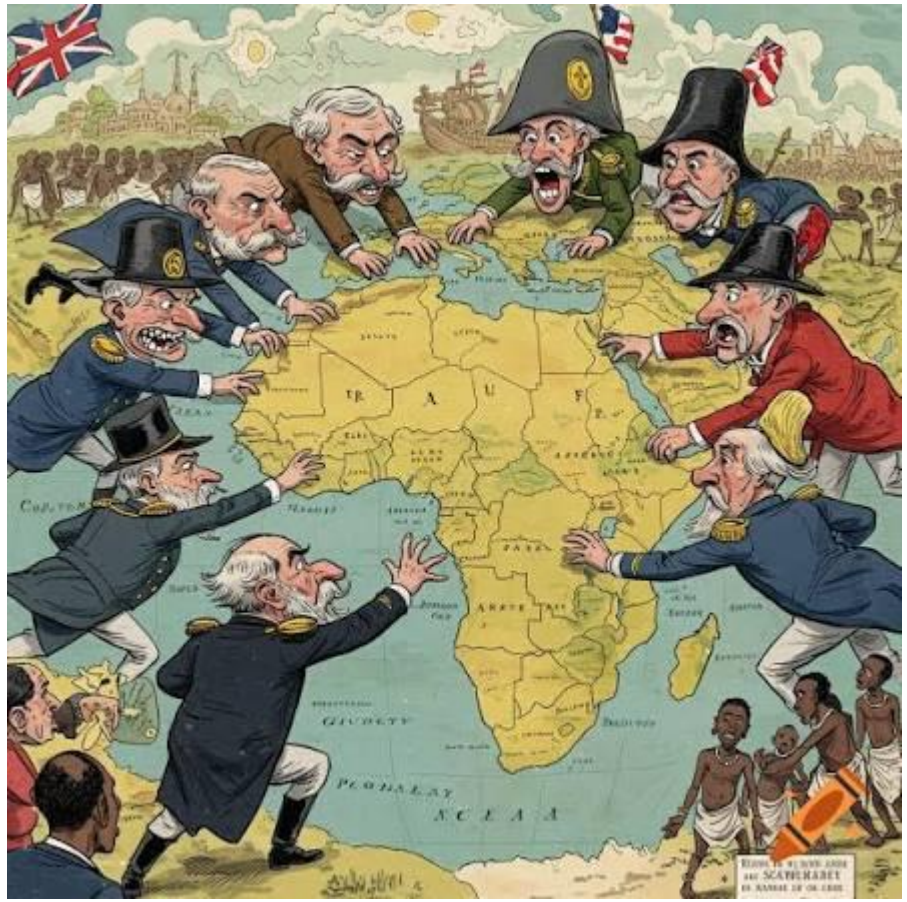
Oral vs written • What counts as philosophy?

African philosophy: challenges

- Knowledge preserved orally through proverbs, stories, songs, and sages
- Colonialism disrupted transmission and devalued oral forms
- Academic philosophy long dismissed African thought as outside logos
- If knowledge is oral, is it less authoritative?



Colonial erasure and division



1415

Portuguese seizure of Ceuta marks an early phase of European colonial expansion.

LATE 19TH / EARLY 20TH C.

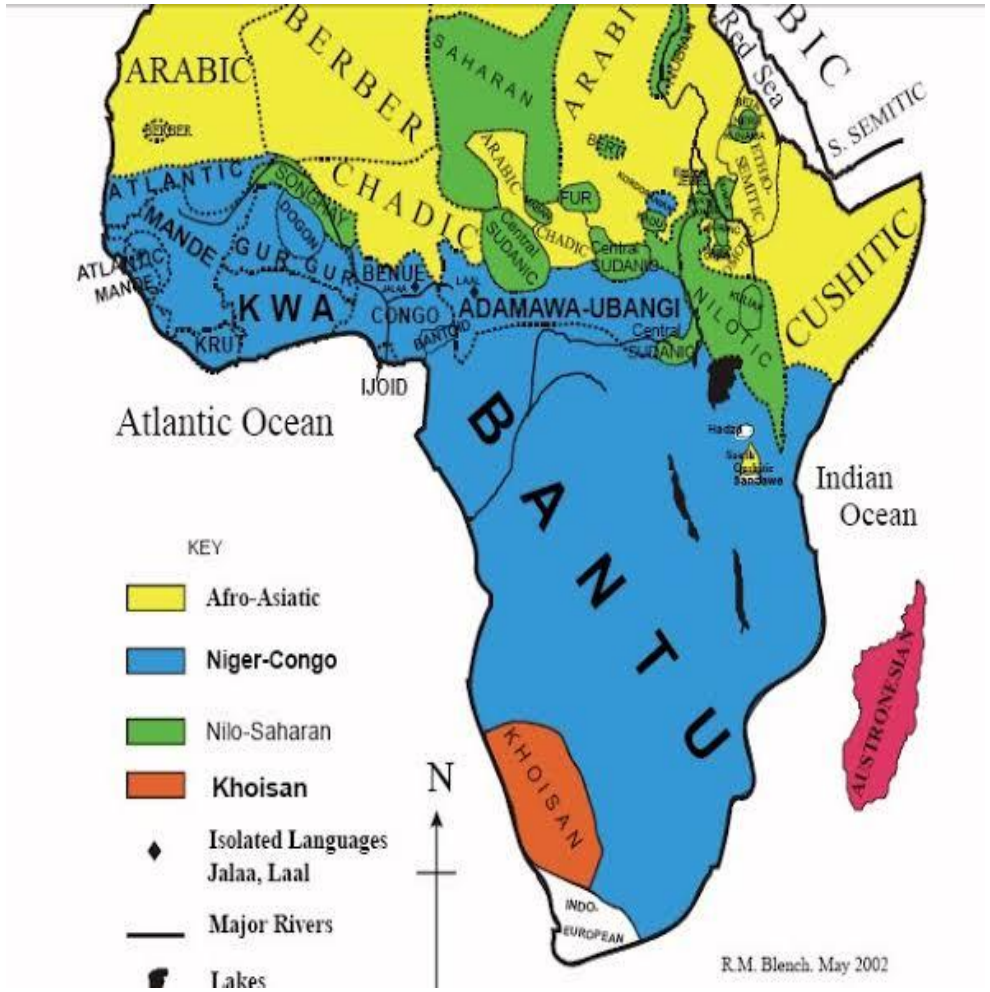
The “Scramble for Africa” accelerates territorial conquest.

BERLIN CONFERENCE

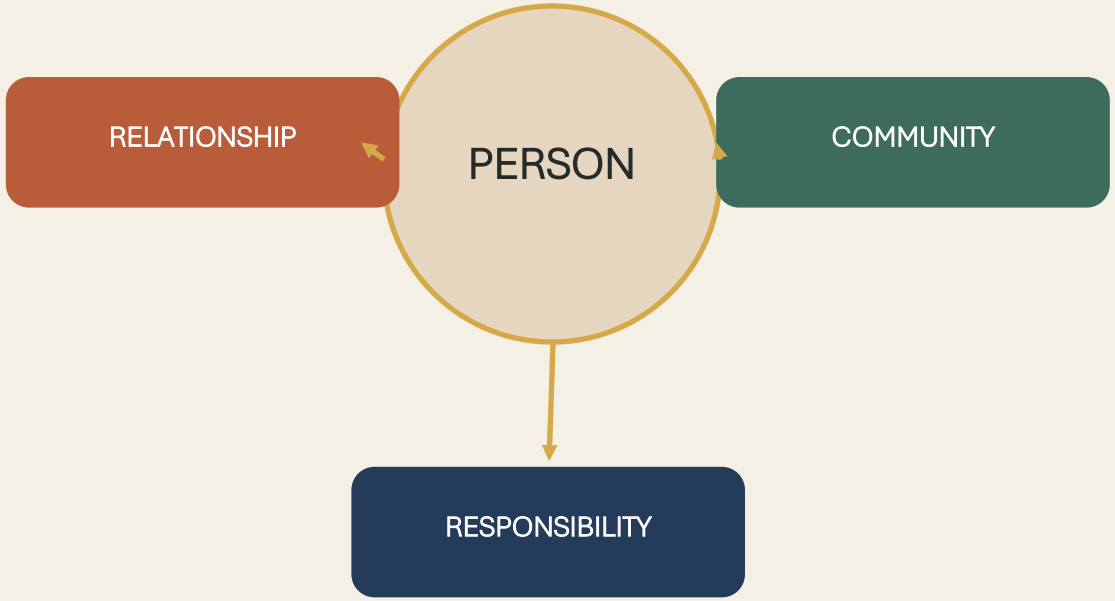
Borders are drawn with little regard for existing peoples and polities.

Representation follows power: “without history” becomes “without philosophy.”

Bantu philosophy and ubuntu



“I am because we are”



Bantu names a broad family of languages and cultures, not a single people.

Yoruba philosophy: Ọ̀rúnmìlà and Socrates

Topic	Socrates's Quote	Ọ̀rúnmìlà's Quote
The nature of truth	"But the highest truth is that which is eternal and unchangeable."	"Truth is what the Great Invisible God uses in organizing the world. . . . Truth is the Word that can never be corrupted."
The limits of human knowledge	"And I am called wise for my hearers always imagine that I myself possess the wisdom which I find wanting in others. But the truth is, O men of Athens, that God only is wise. . . . And so I go about the world, obedient to the God."	"When they turned to me and said: 'Bàbá, we now accept that you are the only one who knows the end of everything,' I retorted, 'I myself do not know these things.' For instruction on this matter, you have to go to God through divination, for He alone is the possessor of that sort of wisdom."
Good and bad	"And are not all things either good or evil, or intermediate and indifferent?"	"Tribulation does not come without its good aspects. The positive and the negative constitute an inseparable pair."
Human nature	"No man voluntarily pursues evil, or that which he thinks to be evil. To prefer evil to good is not in human nature."	"No one who knows that the result of honesty is always positive would choose wickedness when s/he is aware that it has a negative reward."

Table 3.1 Olúwọ́lé's Comparison of Socrates's and Ọ̀rúnmìlà's Ideas. (source: Olúwọ́lé 2015)

Both emphasize wisdom and virtue, but they organize difference in contrasting ways.


Ọrọ Iṣiti #5
Prof Sophie Oluwole

Sage philosophy in Kenya

HENRY ODERA ORUKA

1944–1995



FIELDWORK	INTERVIEW	ANALYSIS	CLAIM
Identify sages in communities	Record reflective arguments	Distinguish critical thought from received belief	Philosophy is not unique to literate societies

Philosophy lives in oral dialogue, not only in books.

Oral transmission is a philosophical issue

AFRICA



Oral traditions marginalized by colonialism

INDIA



Vedas transmitted orally as shruti

BUDDHISM + JAINISM



Teachings begin orally, then form canons

Why is orality valued differently in different contexts?

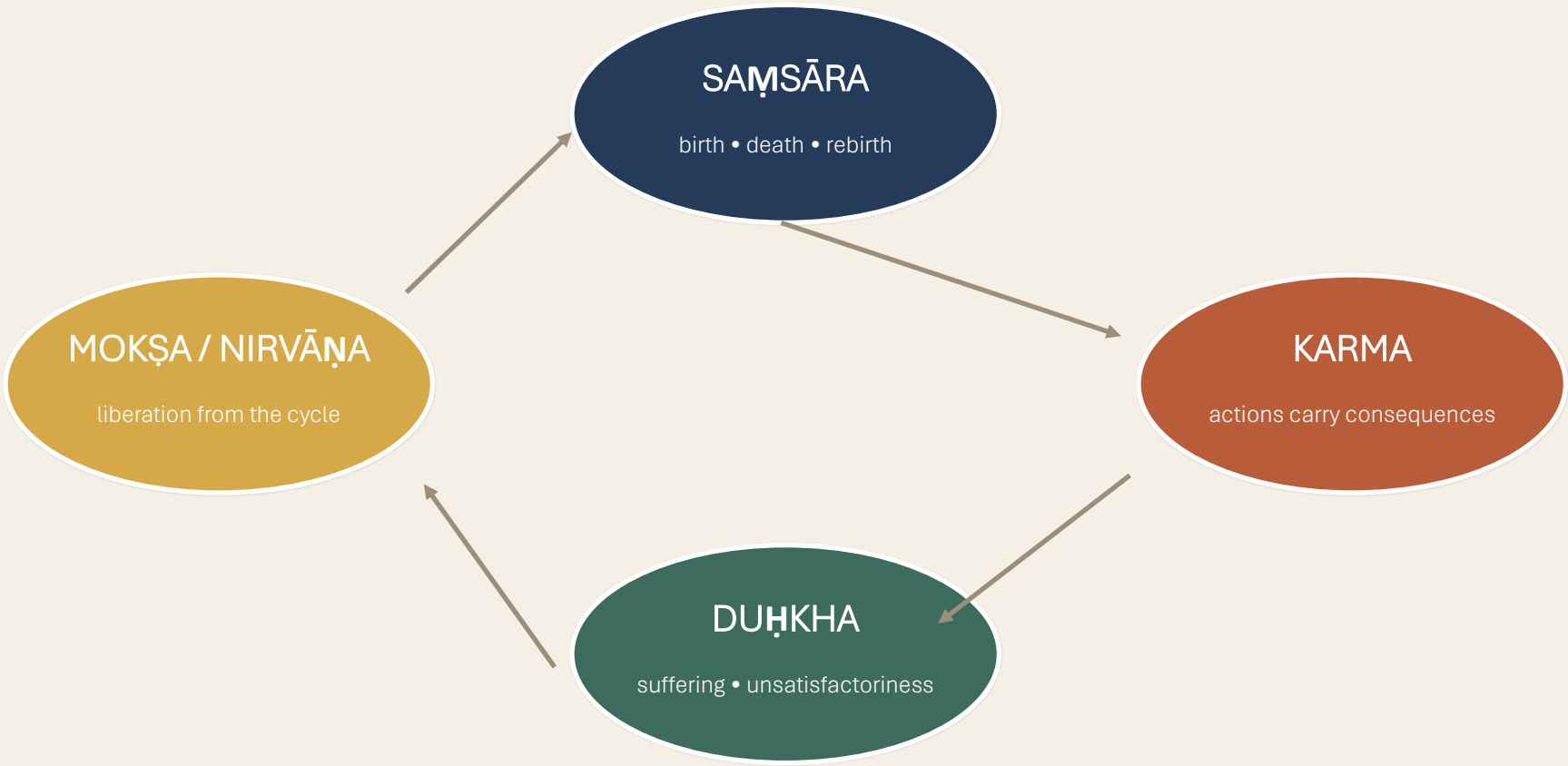
Vedas: shruti, “that which is heard”



- Four Vedas: Rig, Sāma, Yajur, Atharva
- Composed approximately 1500–900 BCE
- Transmitted orally for centuries and regarded as inspired
- Upaniṣads develop the philosophical inquiry, circa 800–200 BCE

ORALITY ≠ LACK OF DISCIPLINE

A problematic



The philosophical task: diagnose bondage and articulate a path to liberation.

Three contrasting responses



This Photo by Unknown Author is licensed under CC BY-SA

VEDĀNTA

True self?
Ātman / Brahman

BUDDHISM

Permanent self?
No: anattā

JAINISM

Soul?
Jīva in living beings

Upanishadic / Vedāntic Hinduism

SELF

True self; in some schools identical with Brahman

AUTHORITY

Vedic and orthodox



KARMA

Moral consequences affect rebirth

PATH

Knowledge, discipline, devotion

Buddhism: no permanent self

ANATTĀ

No unchanging, permanent soul

IMPERMANENCE

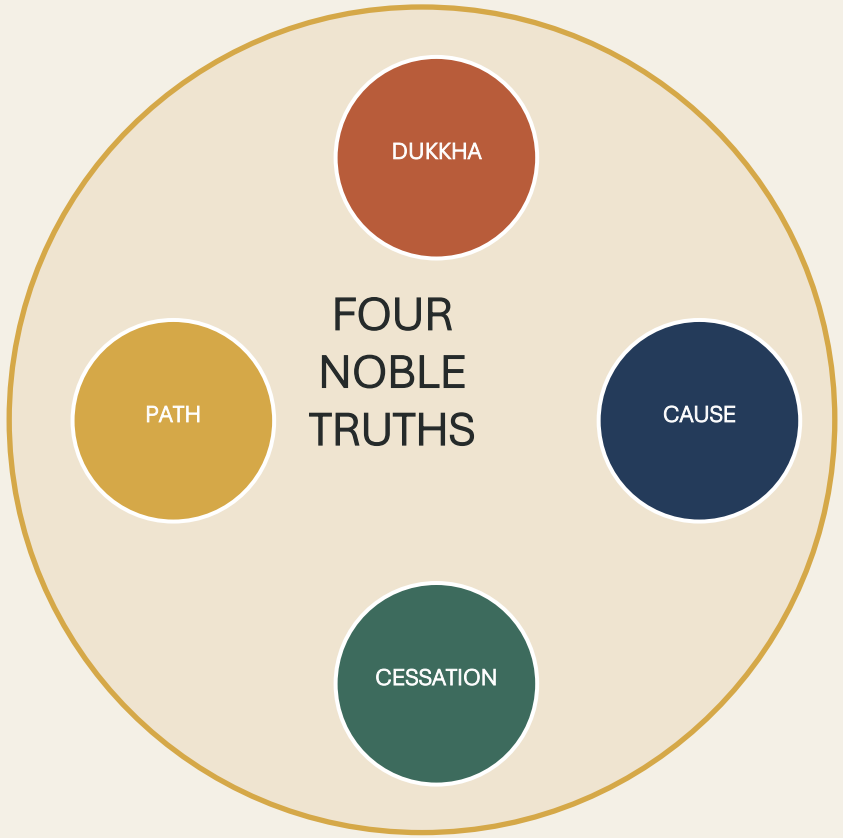
all conditioned things change

DUKKHA

clinging produces unsatisfactoriness

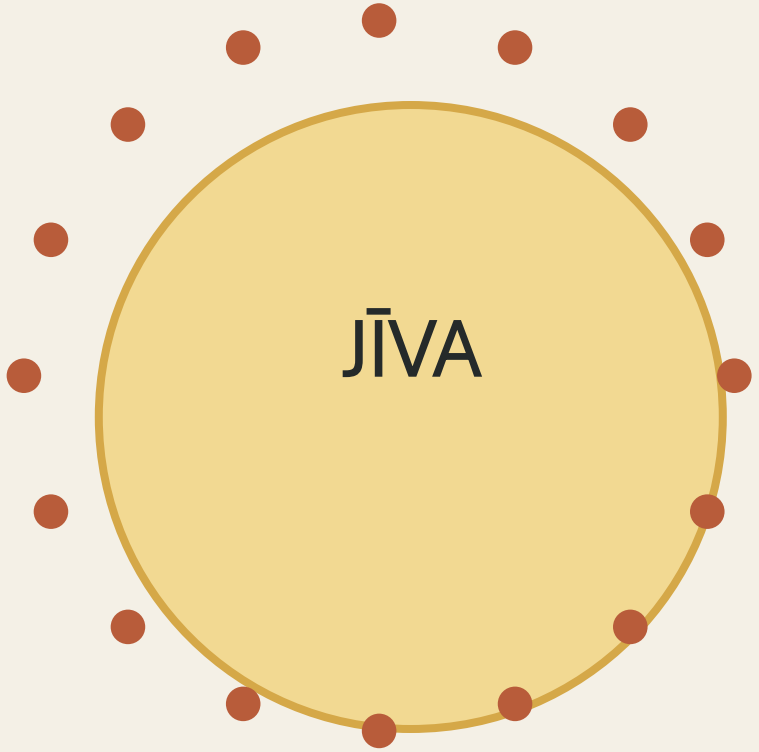
NO-SELF

no permanent essence behind experience



Non-Vedic • heterodox

Jainism: soul bound by karmic matter



Karmic matter binds the eternal soul

AHIṂSĀ

Non-violence is the supreme ethical principle.

MANY LIVING BEINGS

Jīva is present in humans, animals, plants, and more.

AUTHORITY

Non-Vedic and heterodox.

Oral / written and authority in India

TRADITION	TRANSMISSION	STATUS	AUTHORITY
VEDAS	ORAL	CANONICAL	ORTHODOX
BUDDHIST TEACHINGS	ORAL	ALTERNATIVE CANONS	HETERODOX
JAIN TEACHINGS	ORAL	ALTERNATIVE CANONS	HETERODOX

Same feature, different status → Power and canon shape what counts as philosophy

Classical Chinese philosophy: context

WARRING STATES

Political fragmentation
Conflict
Competing visions of order

CENTRAL QUESTION

How can we create order and a good life in a disordered world?

Less emphasis on rebirth • more emphasis on this-worldly harmony

Confucianism: order through relationships



REN
humaneness

LI
ritual • propriety

- Family
- Filial piety
- Education
- Moral rulers

Philosophy as practical cultivation: learning how to live together well.

Daoism: harmony with the Way

DAO — the Way

WU WEI — effortless action

YIN–YANG — dynamic balance

Critique excessive rules, ambition, and artificiality. Align with nature rather than trying to control it.



Two Chinese responses

CONFUCIANISM

Cultivate virtue
Practice ritual
Honor relationships
Lead by moral example

ORDER THROUGH CULTIVATION

HARMONY

DAOISM

Align with Dao
Practice wu wei
Balance yin and yang
Reduce artificial striving

HARMONY THROUGH ALIGNMENT

Bringing it together

AFRICA

Oral traditions
Colonial erasure
Ubuntu
Yoruba sage philosophy

INDIA

Oral + written
Vedic authority
Hindu, Buddhist, Jain
Liberation from saṃsāra

CHINA

This-worldly order
Confucian cultivation
Daoist alignment
Harmony

What counts as philosophy, and where does it live?



The pharaoh Akhenaten, his wife Nefertiti, and their children are blessed by the god Aten, represented by the sun. The Egyptian conception of Aten as the source of all that existed was influential in the metaphysics embraced by the Greeks. (credit: modification of work "Akhenaten, Nefertiti, and the Royal Princesses Blessed by the Aten" by MCAD Library/Flickr, CC BY 2.0)

4.1 Historiography and the History of Philosophy

Presentist Approach

- A **presentist approach** to the history of philosophy examines philosophical texts for the arguments they contain and judges whether their conclusions remain relevant for philosophical concerns today.
- The main limitation to this approach is that it neglects the various contexts in which past philosophers lived and worked.
- In addition to reading philosophical texts too narrowly, the exclusive focus on arguments has been criticized for yielding a profoundly ahistorical understanding of the development of philosophy.



4.1 Historiography and the History of Philosophy

Contextualist Approach

- The **contextualist approach** attempts to understand historical philosophy on its own terms, using concepts and ideas that would have been appropriate to the time period in which they were written.
- There is a danger that contextualist historians might become interested in the history of philosophy for history's sake, ignoring the instrumental value of historical philosophy for contemporary philosophers.



4.1 Historiography and the History of Philosophy

Hermeneutic Approach

- The **hermeneutic approach** takes the historical context of a text seriously, but it also recognizes that a person's interpretation of history is conditioned by their contemporary context.
- A hermeneutic approach to philosophy can fall prey to a tendency to think about history as culminating in the present.



4.2 Classical Philosophy

Egyptian Origins of Classical Philosophy

- The understanding that the roots of classical thought lie in Egypt is as old as the ancient Greeks themselves.
- In *The Histories of Herodotus*, the ancient Greek historian Herodotus traces Greek beliefs about the gods, religious practices, and understanding of the natural world to Egypt.
- Plato spent 13 years in Heliopolis, and Pythagoras studied mathematics in Heliopolis for over two decades (Boas 1948).



4.2 Classical Philosophy

Egyptian and Babylonian Mathematics

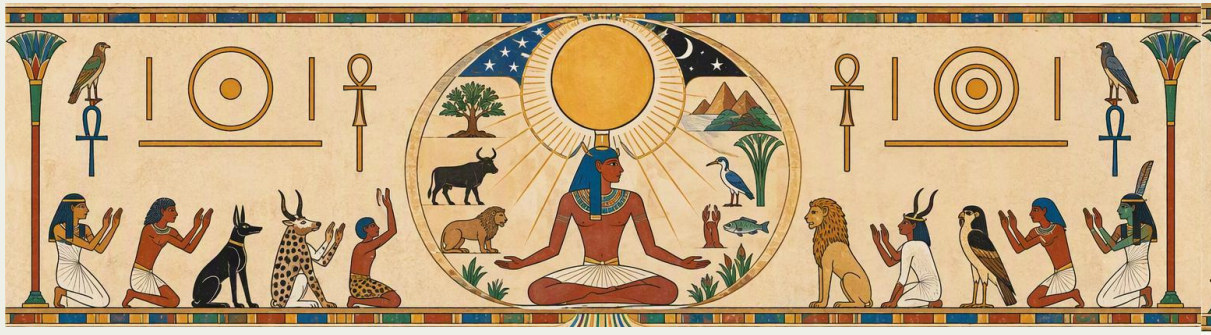
- Pythagoras almost assuredly learned the “Pythagorean” theorem in Egypt.
- A Babylonian clay tablet dating to approximately 1800 BCE, known as Plimpton 322, demonstrates that the Babylonians not only had knowledge of the relationship of the sides and hypotenuses of a right triangle, but also of trigonometric functions (Lamb 2017).
- Pythagoras studied with the priests of Heliopolis more than 1,000 years after these documents were created.



4.2 Classical Philosophy

The Pre-Socratics

- At least a few of the thinkers considered part of this school were contemporaries of Socrates and are mentioned in Plato's dialogues.
  - Foremost among these are the Sophists, traveling teachers of rhetoric who serve as foils for Plato's philosophers.
- Researching the pre-Socratics is difficult because so little of their work has survived.
- Those who think nature consisted of a single substance are called monists, in contrast to pluralists who see it as consisting of multiple substances.





4.2 Classical Philosophy

Prominent Monists

- Thales of Miletus and two of his students, Anaximander and Anaximenes, made up the monist Milesian school.
- Thales proposed that the basic substance of the universe was water.
- Parmenides went so far as to deny the reality of change.
  - Parmenides was the first person to propose that the light from the moon came from the sun and to explain the moon's phases.



4.2 Classical Philosophy

Prominent Pluralists

- Heraclitus held that nothing remains the same and that all is in flux.
- Anaxagoras and Empedocles were substance pluralists who believed that the universe consisted of more than one basic kind of “stuff.”
- The atomists held the view that the basic substance of the universe was tiny, indivisible atoms. For the atomists, all was either atoms or void.



4.2 Classical Philosophy

Pre-Socratic Theology

- The pre-Socratic philosopher Pythagoras and his followers comprised a rational yet mystical sect of learned men.
- A reminder of their mathematical legacy can be found in the Pythagorean theorem.
- Xenophanes rejected the traditional accounts of the Olympian gods. He sought a rational basis of religion and was among the first to claim that the gods are actually projections of the human mind.



4.2 Classical Philosophy

Socrates and Plato

- As Socrates never wrote anything, he is remembered today because thinkers like Plato featured him in their writings.
- The early dialogues feature a skeptical Socrates who refuses to advance any doctrines of his own. Instead, he questions his interlocutors until they despair of finding the truth at all.



4.2 Classical Philosophy

Socrates and Plato

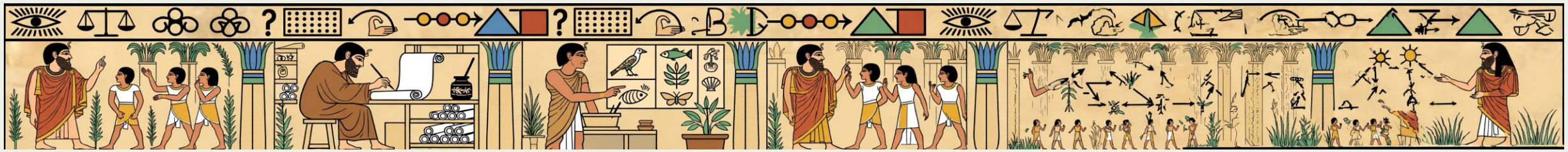
- Gradually, Plato has Socrates give voice to more positive doctrines.
- These include what comes to be known as the **theory of the Forms**, a metaphysical doctrine that holds that every particular thing that exists participates in an immaterial form or essence that gives this thing its identity.



4.2 Classical Philosophy

Aristotle

- Aristotle's works ranged widely across all the main areas of philosophy, including logic, metaphysics, and ethics.
- In addition, he investigated **natural philosophy**, the fields of study that eventually gave rise to science.
- At the center of Aristotle's work was his doctrine of the four causes. He believed that the nature of any single thing could be understood by answering four basic questions: "What's it made of?" (material cause), "What shape does it have?" (formal cause), "What agent gave it this form?" (efficient cause), and finally, "What is its end goal?" (final cause).



4.2 Classical Philosophy

Epicureans

- The Epicureans believed that all knowledge was derived from sense experience.
- The two principal Greek Epicureans were Epicurus himself and his Roman disciple Lucretius.
- Epicurus believed that one should strive to minimize their dependence upon others while limiting the pain in their lives.
- Epicurus and Lucretius argued that people must overcome the fear of death if they were going to live happy lives.



4.2 Classical Philosophy

Roman Philosophy

- Roman philosophical schools were thoroughly grounded in Greek philosophy.
- Many Roman philosophers chose to write in Greek rather than Latin.



4.2 Classical Philosophy

Rhetoric and Persuasion in Politics

- Aristotle wrote a text called *Rhetoric* in which he sought to analyze rhetoric as the counterpart to philosophy.
- Plato generally sought to keep philosophy distinct from politics and was concerned about the messiness of democratic politics in particular.
- In Rome, examples of philosophers who were also statesmen include Cicero and Marcus Aurelius.



4.2 Classical Philosophy

Stoicism

- The Roman Stoics proposed four core virtues: courage, justice, temperance, and wisdom.
- The Stoics prized rational self-control above everything else.
- The Stoic god is a material entity who exists in nature and meticulously manages it, the material first cause of the universe.

4.2 Classical Philosophy

Academic Skepticism

- Academic skeptics opposed the Stoic claims that sense impressions could yield true knowledge, holding instead that knowledge is impossible.
- Instead of knowledge, academic skeptics articulated the idea of degrees of belief.
- The philosopher, orator, and statesman Cicero was the most prominent of the academic skeptics.

4.2 Classical Philosophy

Neoplatonism

- Plotinus led a revival of Plato's thought in the late Roman Empire.
- Neoplatonism arose during a time of cultural ferment in the Roman Empire, incorporating ideas borrowed from sources such as Judaism and early Christianity.
- The key metaphysical problem in Neoplatonism was accounting for how a perfect God could create a universe that was manifestly imperfect.

4.3 Jewish, Christian, and Islamic Philosophy

Defining Jewish, Christian, and Islamic Philosophy

- Early Jewish and Christian scholars incorporated ideas of classical Greek and Roman philosophy into their theological studies.
- As Arab conquerors and traders expanded into the Middle East and Africa, the Muslim world also came into contact with classical philosophy and the natural sciences, adopting and advancing many key ideas.

4.3 Jewish, Christian, and Islamic Philosophy

Early Jewish Philosophy - Philo of Alexandria

- Philo of Alexandria published both his philosophical treatises and his personal accounts of his political experiences.
- His work represents the first systematic attempt to make use of ideas developed by Plato and other Greek philosophers to explain and justify Jewish scripture.
- In reconciling Jewish and Greek doctrines of creation, Philo identifies Plato's forms as **logos**, or the thoughts of God.

4.3 Jewish, Christian, and Islamic Philosophy

Early Jewish Ethics and Metaphysics

- Following the Roman empire's destruction of Jerusalem and the Kingdom of Judah in 70 CE, the Sanhedrin began transcribing the oral traditions so as not to lose them.
- These writings would later become the Talmud. Among these writings is the text *Ethics of Our Fathers*, which provides a moral guide to everyday life.
- Later, Jewish scholars also began to explore metaphysics, culminating in the Kabbalah, which examines the relationship between God and the finite world.
- The continuation of Philo's work fell to a subgroup of Jews whose new religion, Christianity, would be adopted by Rome.

4.3 Jewish, Christian, and Islamic Philosophy

Early Christian Philosophy - Augustine

- Augustine was one of the most influential philosophers and theologians of late antiquity.
- In his *Confessions*, he used his own life and the story of his initially reluctant turn to Christianity as an allegory for understanding God's universe and humanity's place within it.
- Augustine famously posits a theory of time that holds that humans experience the temporal present in three different ways: the present anticipates the future and bleeds into the recent past.
- Augustine not only articulated Christian doctrine that shaped medieval European philosophy for centuries to come, he raised questions that are still being pondered today.

4.3 Jewish, Christian, and Islamic Philosophy

Boethius

- Like Augustine, Boethius was a philosopher who straddled the late Roman and Christian worlds.
- Boethius is best known for his work *The Consolation of Philosophy.*
- Extremely popular throughout the Middle Ages and Renaissance (Marenbon 2020), his work and influence exemplify how Catholicism incorporated classical philosophy into its worldview.

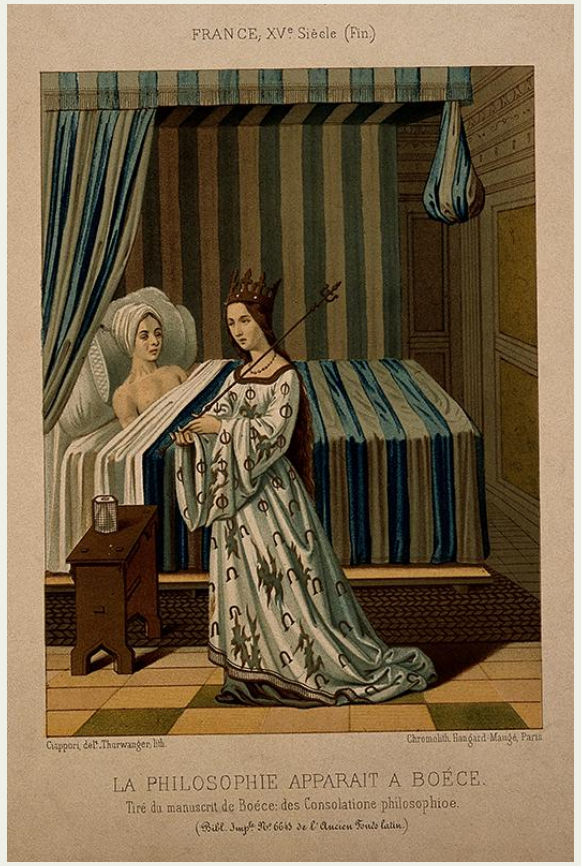


Figure 4.8 In this copy of a 15th-century painting, Lady Philosophy consoles Boethius as he faces death.
(credit: "The Figure of Philosophy Appearing to Boethius" by Wellcome Collection/Wikimedia, CC BY 4.0)

4.3 Jewish, Christian, and Islamic Philosophy

Anselm

- Anselm served as Bishop of Canterbury and sought to extend the reach of Christianity into the British Isles.
- Philosophically, he is best known for a proof for the existence of God, which he elaborated in his written meditation the *Proslogion.*
- Anselm believed that a rational system of thought reflects the rationality inherent in the universe, and that reason and logic can lead people to God.

4.3 Jewish, Christian, and Islamic Philosophy

Islamic Philosophy

- In the early years of Islam, theologians prohibited the teaching of Aristotle and other Greek philosophers. This restriction began to give way in the eighth century CE, which led to the flourishing of philosophy in the Islamic world.



Figure 4.9 This statue of Ibn Sina in Tehran, Iran, honors this highly influential thinker, who published works in philosophy, medicine, astronomy, alchemy, geography, mathematics, Islamic theology, and poetry.
(credit: "Avicenna - Ibn Sina" by Blondinrikard Fröberg/flickr, CC BY 2.0)

4.3 Jewish, Christian, and Islamic Philosophy

Ibn Sina (Avicenna)

- Ibn Sina (called “Avicenna” by Latin writers) was an Islamic philosopher who was influenced by Aristotle. He believed that God created the world necessarily, from all eternity. He also believed that the soul is immortal.
- Ibn Sina wrote on logic and the scientific method, talking of various types of proposition and various types of argument.
- He pointed out that to make a generalization based on properly tested data, you need to have a cause.
- One can discover causes by this type of argument:
  - Fire burns flesh.
  - Fire is hot.
  - Therefore, hot things burn flesh.

4.3 Jewish, Christian, and Islamic Philosophy

Ibn Rushd (Averroes)

- Ibn Rushd (called “Averroes” by Latin writers) was another Islamic philosopher influenced by Aristotle.
- He argued for the existence of God in at least two ways. He thought that universe was too fine-tuned to human existence to be that way without being designed. This argument is still used today.
- He also argued that living things were too complex to have arisen without being designed. Today this view is taken up by advocates of “intelligent design.”
- Ibn Rushd believed that God created the world from preexisting matter.
- Ibn Rushd believed that souls are immortal, but not individually: he believed that at death we all become part of one soul.

4.3 Jewish, Christian, and Islamic Philosophy

Al-Ghazâlî, the Incoherence of the Philosophers

- Al-Ghazâlî was one of the most prominent Sunni Muslim theologians and philosophers.
- Writing in a period after the initial establishment of the Sunni sect, he sought to refute various challenges to its teachings from both Shi`ite religious scholars and philosophers.

4.3 Jewish, Christian, and Islamic Philosophy

Late Medieval Philosophy in Christian Europe

Bonaventure

- St. Bonaventure was one of the Scholastics: Catholic philosophers of the late Middle Ages.
- He was influenced by Augustine and somewhat by Aristotle.
- He thought “knowledge of divine truth is impressed on every soul,” and God is the source of all knowledge. This is called his “illumination” theory.
- He believed that the soul is a thing distinct from the body, with its own kind of matter, and consequently is immortal.

4.3 Jewish, Christian, and Islamic Philosophy

Thomas Aquinas

- St. Thomas Aquinas was the most influential of the Scholastics.
- He agreed with Aristotle in thinking that the soul is the form of the body but disagreed with Aristotle in thinking that the human soul can separate from the body and so is immortal.
- He attempted to prove the existence of God in his “Five Ways”: (1) from motion, (2) from causes, (3) from necessary being, (4) from degrees of being, and (5) from design in nature.
- He adopted Aristotle’s virtue ethics to Christian purposes.

4.3 Jewish, Christian, and Islamic Philosophy

Moses Maimonides

- Moses ben Maimon, or Maimonides, was a physician, Torah scholar, and astronomer in addition to being a philosopher.
- Maimonides proposed that philosophers developed doctrines that do not follow from objective evidence and reason, whereas theologians erroneously interpreted religious texts literally (Bokser 1947).
- Maimonides presents a demythologized conception of the divine.



Figure 4.10 Although deeply religious, Maimonides opposed both literal interpretations of the Bible and anthropomorphized images of God, arguing that God cannot be imagined or even assigned attributes. This statue of Maimonides stands in his birthplace of Cordoba, Spain.
(credit: "Maimónides" by Marco Chiesa/flickr, CC BY 2.0)

4.3 Jewish, Christian, and Islamic Philosophy

Levi ben Gershom (Gersonides)

- Like Maimonides, Gersonides sought to demonstrate the compatibility between Jewish faith and reason.
- His most well-known work, *Wars of the Lord,* takes up the problem of the relationship between Torah, or Jewish scripture, on the one hand and reason on the other.

4.3 Jewish, Christian, and Islamic Philosophy

The Rise of Reason in the Early Modern Era

- In philosophy, the early modern era is delineated by the rapid advancement of natural philosophy, which in turn sparked the scientific revolution.

Nicolaus Copernicus

- Nicolaus Copernicus, born in Poland and raised by his uncle who was a bishop in the Catholic Church.
- At the time, the Catholic Church espoused geocentric model of the solar system, in which the sun and the planets revolve around Earth. Copernicus's mathematical analysis of the astronomical data indicated that Earth and other planets revolved around the sun.

4.3 Jewish, Christian, and Islamic Philosophy

Zera Yacob

- Zera Yacob studied Christian, Jewish, and Islamic thought.
- Although deeply religious, Yacob argued against the supremacy of one religion over another. Rather, he counseled that people must rely on reason to evaluate religious tracts and traditions—and in this way, reach God.
- In Yacob's view, the stories of miracles arose from false human understanding.

This OpenStax ancillary resource is © Rice University under a CC BY-NC-SA 4.0 International license; it may be reproduced or modified for noncommercial purposes only but must be attributed to OpenStax, Rice University and any changes must be noted. Any adaptation must be shared under the same type of license.